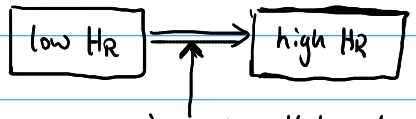
CHE 318 L28

Humidification process: dynamic evaporation

From last lecture we know that vapour press of H_2O varies by T ⇒ relative humidity changes a lot!

The reversal process: evaporation takes away heat can also be explained by the enthalpy concept.



increase enthalpy / additional heat moves into air

because the heat is associated with water, water temperature will drop

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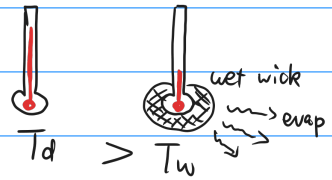
- Final concept we'll introduce is the "wet-bulb temp" T_w

In midterm we talk about interfacial temp ↓ due to evaporation

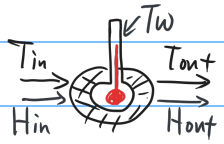
at steady state, when the interface in dry air is T_d, given the S.S. humidity, we will measure a decreased Temp T_w

$T_w < T_d$



(Conceptual experiment T_w is universal lower bound for humidification process).

Wet bulb setup analysis



At steady state (a big room) with convection

① $T_{in} = T_{out}$, but $T_w < T_{in}$

② Presumably $H_{out} > H_{in}$

③ What's happening? $H_y = C_s(T - T_0) + H\lambda_0$

↑ enthalpy

if H humidify changes, the latent heat flows from the thermometer → air flow but not essentially change T

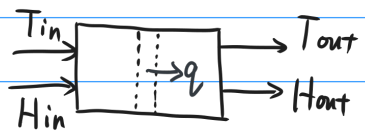
Useful concepts from last lecture

$$h_y = (1.005 + 1.88\,H)(T - T_0) + 2501.4\,H$$

enthalpy 1 kg dry air

abs humidity measured in $\frac{\text{kg } H_2O}{\text{kg air}}$

Wet-bulb experiment: flows (needs some basic heat transfer)



For this control volume, write a heat balance eqn.

Heat flux $\vec{q}$ from L→R, which balances

$$Q_{in} - Q_{out} = \text{Latent}_{in} - \text{Latent}_{out}$$

$$q = m_A \cdot \lambda_w \cdot A \cdot N_A$$

- m_A mol. weight of H_2O (18 kg/kgmol)
- λ_w latent heat of evaporation @ T_w (kJ/kg H_2O)
- N_A total mass flux of H_2O kg mol/s

How do we get N_A? (Mass Transfer coeff)

$$N_A = \frac{k_y'}{y_{Bm}}(y_w - y) \Rightarrow k_y'/y_{Bm} = k_y \quad y_{Bm} \simeq 1$$

y_w: mol. frac at bulb interface (T_w)
y: molar fraction in bulk gas

For humidity, need to convert y from H

$$y = \frac{[\text{mol of } H_2O]}{[\text{mol of } H_2O] + [\text{mol. of air}]} = \frac{H/m_A}{1/m_B + H/m_A}$$

$$\simeq \frac{m_B}{m_A} H$$

(H not R.H. so usually small)

Rewrite $q = m_A \lambda_w A N_A$

$$= \cancel{m_A} \cdot \lambda_w \cdot A \cdot \frac{m_B}{\cancel{m_A}} \cdot k_y \cdot (H_w - H)$$

L.H.S

Fourier's law

effective k in stagnant B

$$q = h(T - T_w) \cdot A$$

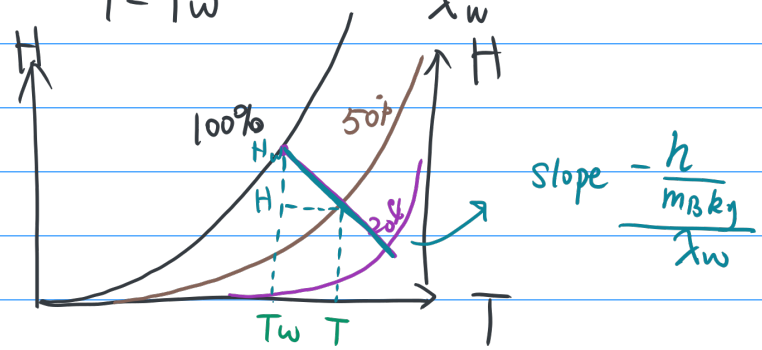
heat transfer coeff

Rearrange everything

heat transfer

mass transfer

$$\frac{H - H_w}{T - T_w} = -\frac{h / m_B k_y}{\lambda_w}$$



slope $-\dfrac{h}{m_B k_y / \lambda_w}$

H_p not H_R

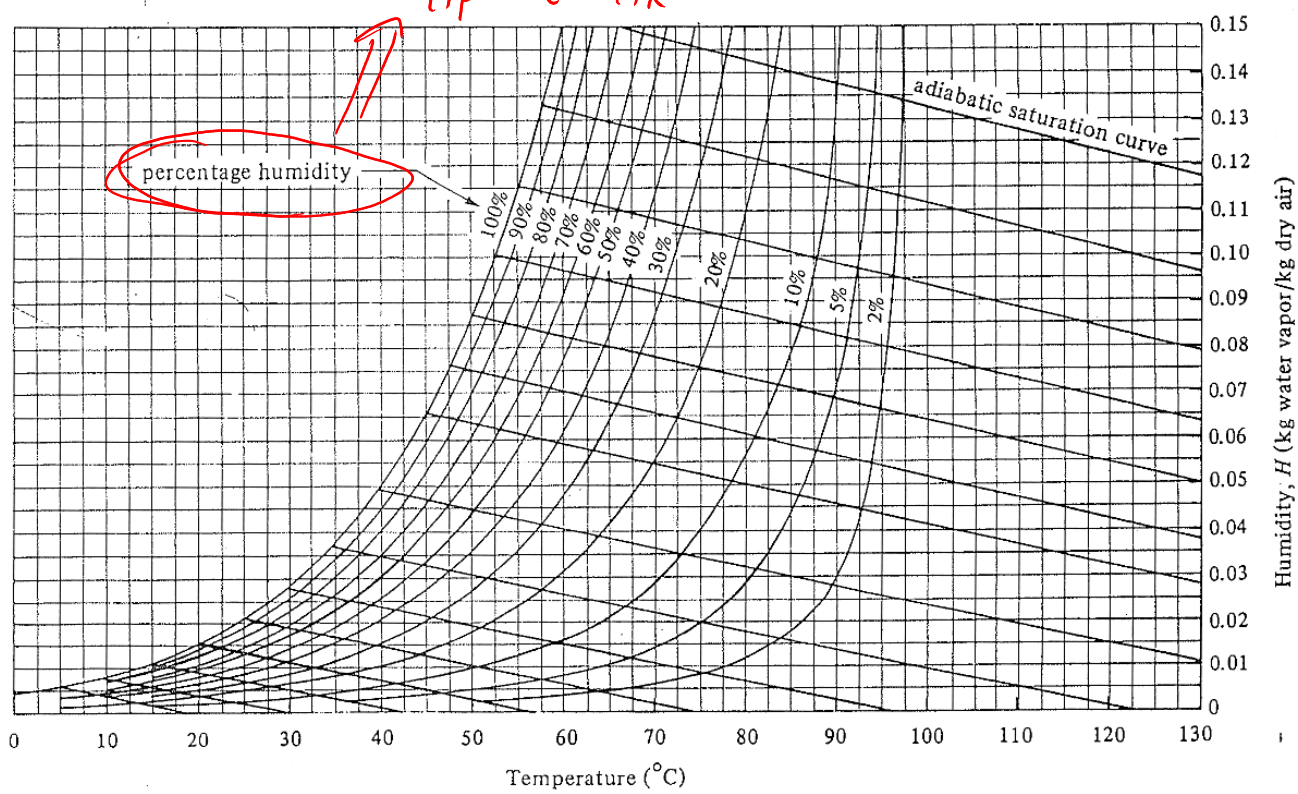


FIGURE 9.3-2. *Humidity chart for mixtures of air and water vapor at a total pressure of 101.325 kPa (760 mm Hg). (From R. E. Treybal, Mass-Transfer Operations, 3rd ed. New York: McGraw-Hill Book Company, 1980. With permission.)*

Look at the equation again → heat transfer coefficient

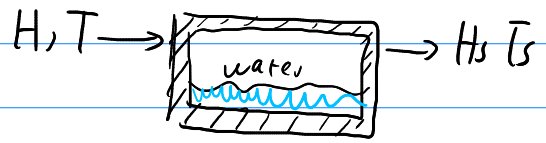
$$\frac{H - H_w}{T - T_w} = -\frac{h/(M_B k_y)}{\lambda_w}$$

↑ w: wet-bulb

Ratio $\frac{h}{M_B k_y}$ = psychrometric ratio (psychro- = cold, metric = measure) measure wet bulb

In the adiabatic (no external heat exchange) process

Heat balance:



saturated temperature T_s

Coincidentally, $\frac{h}{M_B k_y} \simeq 0.96 - 1.025$

$C_s \simeq 1.005 + 1.88 \cdot H$

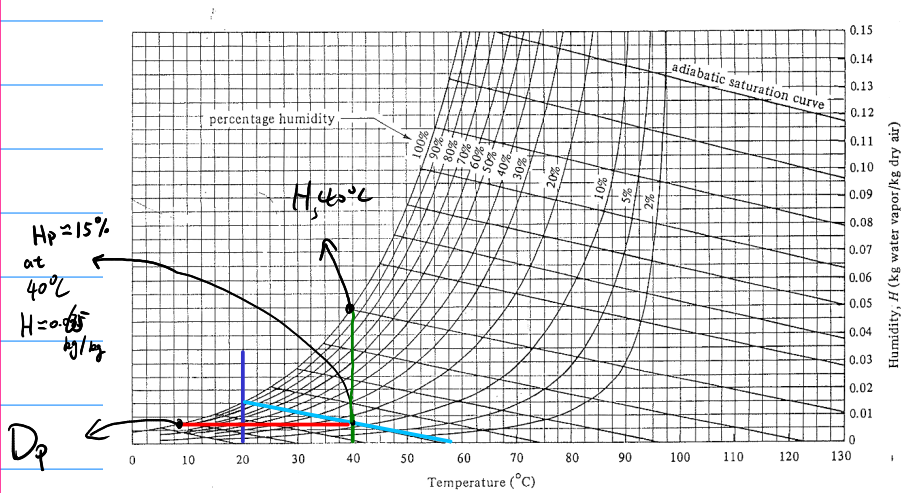
Use the "adiabatic line" ↓ T_w

Let's use the humidity chart for some analysis

Humid air at 40°C has a wet bulb temp of 20°C

what is H in current air, and calulate H_p, Dew point, V_H humid heat & enthalpy

$H(20°C, sat) = 0.0147$ kg w/kg air



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Follow the chart

$$\frac{H - H_w}{T - T_w} = -\frac{1.005 + 1.88H}{\lambda_s}$$

or to solve

$$C_s(T - T_s) + H\lambda_s = C_s(T_s - T_s) + H_s\lambda_s$$

$H(40°C) = 0.0065$

$$\frac{H - H_w}{T - T_w} = -\frac{1.005 + 1.88H}{2501.4}$$

Percentage humidity

at 40°C $\quad H(40°C) = 0.049$

$$H_p = \frac{0.0065}{0.049} = 13\%$$

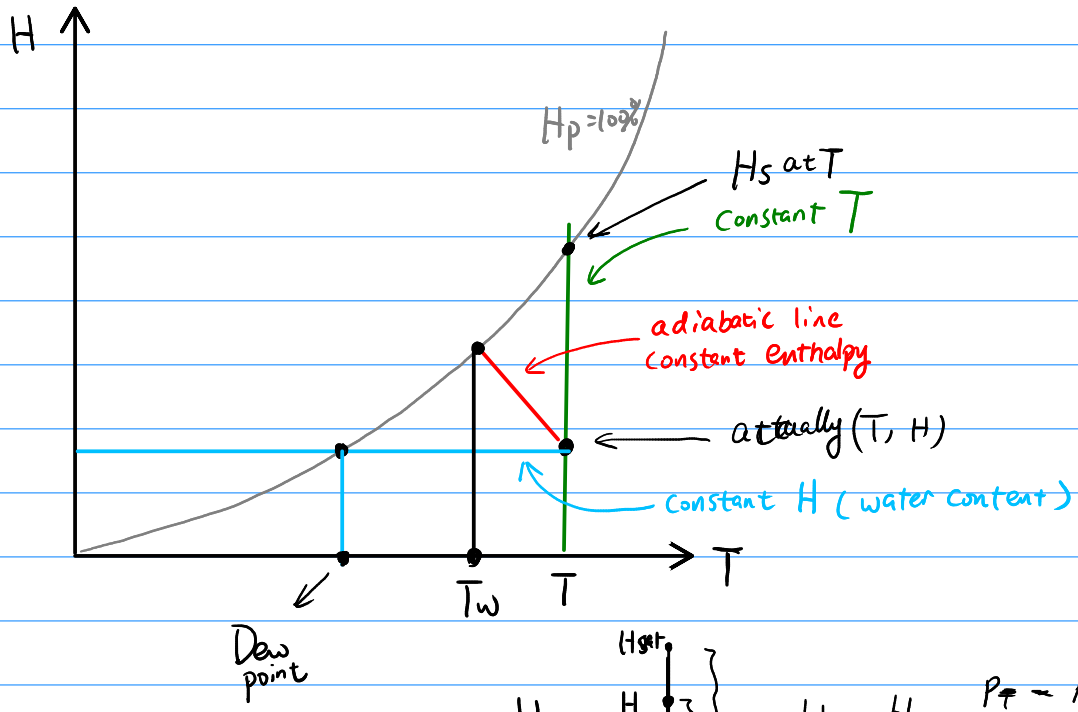
Relative humidity

$$H_R = \frac{P_A}{P_T - P_A} \simeq 14\%$$

Dew point : around 8°C

$V_H \doteq 0.90\ m^3/kg$ dry air $\quad$ (use H at 40°C)

$C_s = 1.005 + 1.88H \doteq 1.02\ kJ/(\text{dry air})/$

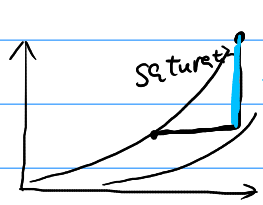


H_p: H_{sat}, H, 0 $\quad H_p = H_R \cdot \dfrac{P_T - P_{vap}}{P_T - P_A}$

H_p slight smaller than H_R

What is the indication of the wet bulb temperature ?

1. cooling of water using dry air is possible (max cooling T_w)
2. vapour pressure difference is driving force (see previous discussion of humid hot)



→ driving force

this is the concept behind cooling tower